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Past is Prologue: Inference from the Cross Section of Returns Around an Event

Contents

1	Big picture	3
2	Data and measurement (key objects)	4
2.1	Sample construction	4
2.2	The eight firm characteristics	4
2.3	Summary statistics from Table 1	4
2.4	Practical choices for the time-series methods	5
3	Models and empirical specifications (all equations + interpretation)	5
3.1	Conceptual framework: event news plus routine news	5
3.2	The standard event-day cross-sectional regression	5
3.3	Why cross-sectional standard errors are too small	6
3.4	Why clustering standard errors does not fix the problem	6
3.5	Time-series benchmark: estimate the event effect relative to ordinary days	7
3.6	Two ways to test significance in the time-series approach	7
3.7	The novel TS-GLS procedure	8
3.8	Potential failure: time-varying routine-news volatility	8
3.9	Why simple panel approaches also fail	9
3.10	Appendix B: general framework	9
4	Identification and instruments (what makes the estimates credible)	10
4.1	Identification idea	10
4.2	Assumptions behind valid inference	10
4.3	Why clustering and factor adjustment have limited identifying value here	10
4.4	Implementation choices as identification choices	10
4.5	External evidence on current practice	11

5	Tables and figures	11
5.1	Figure 1: Daily relationship between returns and investment in 2019	11
5.2	Table 1: Summary statistics	12
5.3	Figure 2: Significance rates based on cross-sectional standard errors	12
5.4	Table 2: Alternative specifications for cross-sectional tests	13
5.5	Table 3: Alternative clustering choices	13
5.6	Figure 3: Significance rates based on TS-OLS and TS-GLS	13
5.7	Figure 4 and Table 4: Detection rates in artificial event periods	14
5.8	Table 5: Robustness of TS-GLS to L and K	14
5.9	Figure 5: Explanatory power of principal components	15
5.10	Table 6 and Figure 6: What the principal components are picking up	16
5.11	Appendix Table A.1: Survey of recent event-study practice	16
5.12	Appendix Figures C.1 and C.2: Choosing the number of PCs	18
6	Short summary	18
7	Conclusion	19
7.1	Core conclusion (what the paper establishes)	19
7.2	Economic intuition	19
7.3	Takeaway	19

1 Big picture

- **Question.** When researchers regress short-horizon stock returns around a quasi-experimental event on a firm characteristic that proxies for exposure, how should they do inference? The paper asks whether the usual cross-sectional regression standard errors are valid, and if not, what should replace them.
- **Core threat.** On the event day, the focal event is not the only thing happening. “Routine news” also arrives, and that routine news can move firms with high and low values of the same characteristic differently. This creates genuine return-characteristic slopes even when the focal event itself has no differential effect.
- **Main theoretical point.** Standard event-day cross-sectional regressions test the wrong null. They test whether the total slope on the event day is zero, not whether the part of that slope attributable to the focal event is zero.
- **Setting.** U.S. common stocks from 1991 to 2021, using CRSP daily returns and Compustat accounting data. The paper studies eight firm characteristics, including classic corporate-finance variables such as size, market-to-book, profitability, and investment, plus four characteristics taken from prominent event-study papers.
- **How severe is the problem?** Very severe. In one-day cross-sectional regressions, the slope on a characteristic is statistically significant at the 1% level on an average of 33.4% of all trading days when using default OLS standard errors. Even with Fama-French 49 industry clustering, the average 1% significance rate is still 18.5%.
- **Why this matters.** A “significant” event-day slope is therefore a very low bar. The event day may look special even when it is not unusual relative to ordinary days.
- **Main solution.** Benchmark the event-day slope against the time-series distribution of the same slope on pre-event days. The paper calls this a time-series approach and studies both an OLS version (TS-OLS) and a new GLS version (TS-GLS).
- **Main inferential result.** If significance is assessed by the empirical CDF of pre-event coefficients rather than a parametric t -test, false positives are brought back to approximately nominal size.
- **Main efficiency result.** The paper’s novel TS-GLS procedure uses principal components of recent return comovements to estimate a daily covariance matrix, then uses that covariance matrix for GLS estimation. This substantially improves power relative to TS-OLS.
- **Quantitative power gain.** At the 5% level, with an artificial one-day effect of 50 basis points per one-standard-deviation change in a characteristic, average detection rises from 72.90% under TS-OLS to 95.46% under TS-GLS. For five-day windows, detection rises from 35.19% to 61.24%.
- **Deeper message.** Clustering and factor adjustment help only a little because the cross-sectional covariance structure of returns is high-dimensional and time-varying. The relevant return comovements are often episode-specific (for example, finance in 2008 and meme stocks in 2021).
- **Why the paper matters.** This is a methodology paper with immediate practical implications for corporate finance and event studies. It provides a simple diagnostic: the event-day slope

should look unusual relative to its own recent history, not merely significant relative to its cross-sectional standard error.

2 Data and measurement (key objects)

2.1 Sample construction

- **Return data.** Daily firm-level stock returns from CRSP.
- **Accounting data.** Annual Compustat data used to build firm characteristics.
- **Sample period.** 1991–2021, containing 7,811 trading days.
- **Universe.** NYSE, AMEX, and NASDAQ common stocks with share codes 10 or 11 and share price above 3 dollars.
- **Exclusions.** Financial and utility firms, defined as firms with one-digit SIC code 4 or 6.
- **Units.** The main unit is the firm-day, though the paper also studies five-day return windows.
- **Sample size.** 21,071,829 firm-days belonging to 11,556 unique firms. The average day has roughly 2,698 firms in the cross section.

2.2 The eight firm characteristics

- **Size.** Log market equity, computed daily from CRSP price times shares outstanding.
- **M/B .** Log market-to-book ratio, computed monthly using market equity at the end of the prior month and book equity from the prior fiscal year.
- **Profit.** Gross profit divided by total assets.
- **Invest.** Capital expenditures divided by total assets.
- **Cash/AT.** Cash and short-term investments divided by total assets.
- **Debt/AT.** Long-term debt plus debt in current liabilities, divided by total assets.
- **Tax Rate.** 100 times taxes paid divided by pre-tax income net of special items, set to 0 when pre-tax income is negative.
- **NY HQ.** Indicator for being headquartered in New York.
- **Outliers.** Continuous characteristics are winsorized at the 1st and 99th percentiles.

2.3 Summary statistics from Table 1

- **Returns.** The mean one-day return is 0.12%, the median is 0.00%, the total standard deviation is 4.00%, and the average within-day cross-sectional standard deviation is 3.82%.
- **Coverage.** Most characteristics have roughly 20–21 million firm-day observations. Tax Rate has lower coverage, about 14.39 million firm-days.

- **Dispersion.** Size has mean 6.14 and within-day standard deviation 1.81; M/B has mean 1.01 and within-day standard deviation 0.96; Profit has mean 0.36; Invest has mean 0.06.
- **Interpretation.** There is abundant cross-sectional variation in every characteristic, so the issue in the paper is not lack of cross-sectional variation. The problem is that ordinary-day news already loads strongly on these characteristics.

2.4 Practical choices for the time-series methods

- **Pre-event window length.** The baseline uses $L = 199$ pre-event days. This is long enough to estimate the benchmark distribution but short enough to reduce concerns about non-stationarity.
- **Why 199?** For the empirical CDF test, $L + 1$ should be divisible by $1/p^*$, so 199 is convenient for 1% and 5% thresholds.
- **Timing of characteristics.** Each characteristic on day t is measured using the most recent information available strictly before day t , which avoids look-ahead bias and is more efficient than freezing characteristics at the earliest pre-event date.
- **TS-GLS covariance estimation.** The baseline uses $T_K = 200$ prior trading days to estimate return principal components and $K = 100$ principal components to build the daily covariance matrix.
- **Robustness grid.** The paper also studies $L \in \{99, 199, 299\}$ and $K \in \{50, 100, 150\}$.

3 Models and empirical specifications (all equations + interpretation)

3.1 Conceptual framework: event news plus routine news

The basic data generating process is

$$r_{i,t} = m_t + (n_t + e\mathbf{1}(t = \tau))(x_{i,t-1} - \bar{x}_{t-1}) + \delta_{i,t}. \quad (1)$$

- **Objects.** $r_{i,t}$ is firm i 's return on day t ; $x_{i,t-1}$ is a pre-determined firm characteristic; e is the differential effect of the focal event; n_t is routine news that also creates differential return responses by x ; m_t is the common mean effect of aggregate news.
- **Economic interpretation.** The event day contains two things at once: the quasi-experimental event and ordinary background news. Both can generate a cross-sectional return slope on x .
- **Why the characteristic is demeaned.** Using $x_{i,t-1} - \bar{x}_{t-1}$ isolates cross-sectional differences in exposure rather than aggregate mean returns.

3.2 The standard event-day cross-sectional regression

Researchers typically estimate

$$r_{i,\tau} = a_\tau + b_\tau x_{i,\tau-1} + \delta_{i,\tau}, \quad (2)$$

where

$$a_\tau = m_\tau - (n_\tau + e)\bar{x}_{\tau-1}, \quad b_\tau = n_\tau + e. \quad (3)$$

- **Interpretation.** The estimated event-day slope combines the focal event and routine news.
- **Important distinction.** Even if $e = 0$, the event-day slope b_τ can be nonzero because n_τ need not be zero.
- **Implication.** A significant estimate of b_τ does not imply a significant estimate of e .

3.3 Why cross-sectional standard errors are too small

The paper's key variance decomposition is

$$\text{Var}(\hat{b}_\tau - e) = \text{Var}(\hat{b}_\tau - b_\tau) + \sigma_n^2. \quad (4)$$

- **Interpretation.** The event-day coefficient has two sources of uncertainty as an estimate of e : usual sampling error and the unobserved realization of routine news.
- **Main insight.** Cross-sectional regression standard errors capture only $\text{Var}(\hat{b}_\tau - b_\tau)$. They ignore σ_n^2 , the variation in routine-news slopes.
- **Why bigger samples do not solve it.** Sampling error shrinks with more firms, but σ_n^2 does not. Even an infinite cross section does not reveal what the event-day slope would have been absent the focal event.

For non-event days $t \neq \tau$, the paper rewrites returns as

$$r_{i,t} = m_t + \nu_{i,t}, \quad (5)$$

with

$$\nu_{i,t} = n_t(x_{i,t-1} - \bar{x}_{t-1}) + \delta_{i,t}, \quad (6)$$

so that the cross-firm covariance induced by routine news is

$$\text{Cov}(\nu_{i,t}, \nu_{j,t}) = (x_{i,t-1} - \bar{x}_{t-1})(x_{j,t-1} - \bar{x}_{t-1})\sigma_n^2. \quad (7)$$

- **Interpretation.** Firms with similar values of x co-move because routine news affects them similarly.
- **Connection to the later GLS procedure.** The covariance matrix of returns contains information about the way routine news loads on characteristics.

3.4 Why clustering standard errors does not fix the problem

- **Paper's point.** Clustering can help when regression *residuals* are correlated, but the confounding-events problem is not fundamentally about slope uncertainty.

- **What actually moves.** The problem is that the entire slope b_τ shifts because of unobserved routine news.
- **Consequence.** Residuals from the event-day cross-sectional regression do not contain information about the missing uncertainty term σ_n^2 .

3.5 Time-series benchmark: estimate the event effect relative to ordinary days

The time-series procedure begins by estimating a daily cross-sectional regression on the event day and on L pre-event days:

$$r_{i,t} = a_t + b_t x_{i,t-1} + \epsilon_{i,t}, \quad t \in \{\tau, \tau - 1, \dots, \tau - L\}. \quad (8)$$

The estimated event-specific effect is

$$\hat{e} = \hat{b}_\tau - \mu(\hat{b}), \quad \mu(\hat{b}) = \frac{1}{L} \sum_{\ell=1}^L \hat{b}_{\tau-\ell}. \quad (9)$$

Under stationarity of the routine-news and sampling-error distributions,

$$\mathbb{E}(\hat{e} - e) = 0, \quad (10)$$

$$\text{Var}(\hat{e} - e) = \text{Var}(\hat{b}_{\tau-\ell}), \quad (11)$$

and

$$\Phi(\hat{e} - e) = \Phi(\hat{b}_{\tau-\ell}). \quad (12)$$

- **Interpretation.** The pre-event slopes provide the empirical distribution of the event-day slope under the null that the focal event has no differential effect.
- **Why it works.** Routine news affects the pre-event days too, so the time-series of ordinary-day slopes automatically includes the volatility that was missing from the event-day cross-sectional standard error.
- **Analogy.** This is close in spirit to Fama-MacBeth: inference comes from the time-series variation in cross-sectional slopes.

3.6 Two ways to test significance in the time-series approach

The parametric version computes

$$\hat{t} \equiv \frac{\hat{e}}{\sqrt{\text{Var}(\hat{e} - e)}} = \frac{\hat{b}_\tau - \mu(\hat{b})}{\sqrt{\text{Var}(\hat{b}_{\tau-\ell})}}. \quad (13)$$

The nonparametric empirical-CDF version computes

$$\hat{p} \equiv \frac{1}{L+1} \left[\sum_{\ell=1}^L \mathbf{1} \left(|\hat{b}_{\tau-\ell} - \mu(\hat{b})| > |\hat{b}_{\tau} - \mu(\hat{b})| \right) \right]. \quad (14)$$

- ***t*-test version.** Efficient and familiar, but relies on the distribution of pre-event slopes being well summarized by a standard *t* reference distribution.
- **CDF version.** Fully empirical and robust to fat tails. This becomes the paper’s preferred inferential procedure because the actual distribution of daily slopes is fat-tailed.
- **Two-tailed nature.** The CDF test compares absolute deviations from the pre-event mean.

3.7 The novel TS-GLS procedure

The paper upgrades Step 1 above from OLS to GLS, using a covariance matrix estimated from recent return comovement. Returns are assumed to follow an APT-style structure:

$$r_{i,t} = \phi_i + \sum_{k=1}^K \lambda_{i,k} f_{k,t} + \epsilon_{i,t}, \quad (15)$$

with idiosyncratic covariance

$$\text{Cov}(\epsilon_{i,t}, \epsilon_{j,t}) = \begin{cases} \sigma_i^2 & \text{when } i = j, \\ 0 & \text{when } i \neq j. \end{cases} \quad (16)$$

This implies the estimated covariance matrix of returns

$$\hat{\Omega}_{r,ij} \equiv \text{Cov}(r_{i,t}, r_{j,t}) = \sum_{k=1}^K \lambda_{i,k} \lambda_{j,k} \text{Var}(f_{k,t}) + \mathbf{1}(i = j) \sigma_i^2. \quad (17)$$

In implementation, the daily slope is estimated by GLS using $\hat{\Omega}_t^{-1}$ as the weighting matrix.

- **Key innovation.** Instead of specifying a small number of ad hoc factors, the paper lets PCA extract the statistically important dimensions of return comovement from the previous 200 trading days.
- **Why demean returns first?** Demeaning by day ensures the covariance matrix is suited to cross-sectional regressions with an intercept.
- **Why GLS helps.** GLS increases efficiency of both event-day and pre-event slope estimates. It does *not* solve the confounding-events problem by itself; the time-series comparison does that. GLS simply makes the benchmarked estimator more powerful.

3.8 Potential failure: time-varying routine-news volatility

- **Concern.** If the distribution of routine news differs sharply between the event period and the pre-event window, then pre-event slopes are not a good counterfactual benchmark.

- **Examples.** A tax event embedded in a broader tax reform package, or a regulation change arriving during broad market turmoil.
- **Diagnostics suggested by the paper.** Examine rolling volatility of the daily cross-sectional slope estimates and carefully study the surrounding event environment. A good quasi-natural experiment should not sit inside a major unrelated structural break.

3.9 Why simple panel approaches also fail

The first panel alternative is

$$r_{i,t} = a + b_1 x_{i,t-1} + b_2 x_{i,t-1} \mathbf{1}(t = \tau) + \epsilon_{i,t}. \quad (18)$$

The second is

$$r_{i,t} = a_t + \sum_{\ell=1}^L b_{\ell} x_{i,t-1} \mathbf{1}(t = \tau - \ell) + b_{\tau} x_{i,t-1} \mathbf{1}(t = \tau) + \epsilon_{i,t}. \quad (19)$$

- **Why they fail.** The first panel regression still loads the event coefficient on both e and event-day routine news. The second is algebraically equivalent to the pure event-day cross-sectional regression in terms of the event coefficient and its standard error.
- **Bottom line.** The issue is not lack of panel structure. The issue is the null being tested.

3.10 Appendix B: general framework

The appendix extends the setup to J characteristics, N -day event windows, nonzero average return-characteristic relations, and general covariance in residuals:

$$r_{i,t} = a_t + \left[\lambda + n_t + \frac{e}{N} \mathbf{1}(t \in \tau) \right]' (x_{i,t-1} - \bar{x}_{t-1}) + \delta_{i,t}. \quad (20)$$

The main appendix result for a standard cross-sectional OLS event-window regression is

$$\mathbb{E}[\hat{b}_{\tau} - e] = N\lambda, \quad (21)$$

$$\text{Var}(\hat{b}_{\tau} - e) = \text{Var}(\hat{b}_{\tau} - b_{\tau}) + N\sigma_n^2. \quad (22)$$

- **Interpretation.** If a characteristic has a nonzero unconditional relation with average returns, the standard event-window slope is biased for the event effect by $N\lambda$.
- **Why the appendix matters.** It shows the paper's critique is not tied to the simplest one-day, one-characteristic case. The problem is generic.

4 Identification and instruments (what makes the estimates credible)

4.1 Identification idea

- **Object of interest.** The event-induced differential slope e .
- **Why identification is hard.** The researcher never observes the counterfactual event-day slope absent the focal event, because routine news is always present.
- **Identification strategy.** Use the time-series distribution of non-event slopes as the counterfactual distribution for the event-day slope under $H_0 : e = 0$.
- **Economic content.** Identification comes from abnormality relative to nearby history, not from the cross-sectional precision of one day's regression.

4.2 Assumptions behind valid inference

- **Stationarity over the benchmark window.** The distribution of routine-news slopes and the distribution of sampling error should be stable over the pre-event window.
- **No contamination of the benchmark.** The pre-event window should not already contain information leakage or related events that mechanically resemble the focal event.
- **Why use pre-event rather than post-event days?** To avoid post-announcement drift and follow-on events that would muddy interpretation.
- **Why the CDF approach is preferred.** The daily-slope distribution is fat-tailed, so parametric t -based critical values are a poor approximation even if the basic stationarity logic is correct.

4.3 Why clustering and factor adjustment have limited identifying value here

- **Clustering.** Clustering changes the estimated sampling variance conditional on regressors, but it does not recover the missing time-series uncertainty in the slope itself.
- **Factor adjustment.** Removing market, Fama-French, or other low-dimensional factors slightly lowers ordinary-day significance rates, but it does not come close to restoring nominal size.
- **Deeper reason.** The covariance structure that matters is high-dimensional and unstable across episodes. The later PCA analysis shows why a few pre-specified factors are not enough.

4.4 Implementation choices as identification choices

- $L = 199$. Balances precision against the risk that the return-characteristic process changes too much over time.
- $K = 100$ and $T_K = 200$. Gives a rich covariance estimate without moving too far into the region where overfitting becomes important.

- **Empirical support for these choices.** Appendix C shows that the gains from additional PCs are substantial at low K , flatten between roughly 40 and 190 PCs, and deteriorate only near the saturation point.

4.5 External evidence on current practice

- **Appendix A survey.** The authors manually verify 81 papers from top finance and accounting journals (2014–2023) that use this type of event-study design.
- **What papers currently do.** 64% use simple or White robust standard errors; 20% use industry clustering; 5% use geography clustering; only 6% use a time-series method.
- **Interpretation.** The inferential mistake diagnosed in this paper is not niche. It is common practice.

5 Tables and figures

5.1 Figure 1: Daily relationship between returns and investment in 2019

Read:

- The figure plots the absolute event-day slope of one-day returns on investment, recentered around the annual average and rescaled to basis points per one standard deviation of investment.
- Three eye-catching 2019 events – the Blood Moon, the March 13 Brexit vote, and the October 11 Phase 1 trade-deal announcement – all produce economically noticeable and conventionally significant slopes.
- But the crucial point is comparative: the relationship between returns and investment is significant at the 5% level on 48.4% of all trading days in 2019, and more than 40% of days have larger magnitudes than each of those highlighted events.
- The figure is the paper’s visual punchline. Event-day significance is common on ordinary days, so the right benchmark is not zero; it is the distribution of ordinary-day slopes.

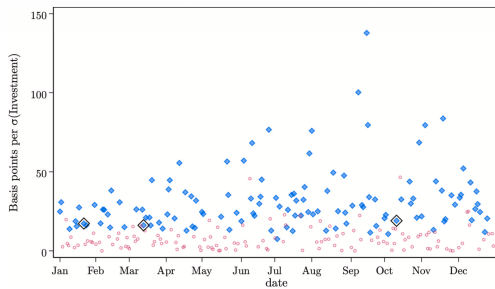


Fig. 1. Daily relationship between returns and investment in 2019. This figure presents daily estimates of the cross-sectional relationship between firm-level returns and *Investment*, where *Investment* is capital expenditures divided by total assets. For each trading day in 2019, we regress one-day returns on *Investment* and compute the absolute value of the difference between the estimated daily coefficient and the average coefficient over the year. We then rescale this difference to basis points per standard deviation of *Investment*. A blue diamond indicates that the raw coefficient is statistically significant at the 5% level based on a t test using White (1980) robust standard errors, while a red circle indicates insignificance at the 5% level. The estimates for January 22 (the day after the Blood Moon, which occurred after trading hours), March 13, and October 11 are enclosed in larger diamonds for emphasis.

Table 1

Summary statistics.

Variable	Firm-Days	Firms	Firms/Day	Mean	Median	σ	Within-day σ
Return (%)	21,071,829	11,556	2698	0.12	0.00	4.00	3.82
Size	21,064,678	11,631	2697	6.14	6.00	1.96	1.81
M/B	20,487,758	11,487	2623	1.01	0.92	0.98	0.96
Profit.	21,084,670	11,650	2699	0.36	0.34	0.32	0.32
Invest.	20,911,833	11,613	2677	0.06	0.04	0.07	0.07
CASH/AT	21,084,729	11,651	2699	0.21	0.12	0.24	0.24
Debt/AT	21,010,836	11,645	2690	0.21	0.16	0.22	0.22
Tax Rate	14,393,605	8110	1843	23.96	23.64	16.65	16.36
NY HQ	20,170,658	11,000	2582	0.07	0.00	0.25	0.25

This table presents summary statistics for the sample of firm-day observations used in our analysis. *Return* is the one-day stock return. *Size* is the natural log of market equity, which is the product of the closing stock price on the previous day and the number of shares outstanding from CRSP. *M/B* is the log of market value of equity at the end of the prior month to book value (Compustat *CEQ*) measured at the end of the prior fiscal year (observations for which book value is zero or negative are excluded). *Profit.* is gross profit (Compustat *GP*) divided by total assets (Compustat *AT*). *Invest.* is capital expenditures (Compustat *CAPX*) divided by total assets. *Cash/AT* is cash and short-term investments (Compustat *CHE*) divided by total assets. *Debt/AT* is the sum of long-term debt (Compustat *DLTT*) and debt in current liabilities (Compustat *DLIC*), divided by total assets. *Tax Rate* is 100 times income taxes paid (Compustat *TXDP*), divided by the difference between pre-tax income (Compustat *PI*) and special items (Compustat *SPI*), set to 0 if $PI - SPI < 0$. *NY HQ* is an indicator variable equal to 1 if a firm is headquartered in New York (Compustat *STATE* equal to "NY") and 0 otherwise. Our sample is the 7,811 trading days in 1991 through 2021.

5.2 Table 1: Summary statistics

Read:

- The sample is huge: more than 21 million firm-days and over 11,500 firms.
- The paper has plenty of cross-sectional variation in the characteristics, so the inferential pathologies are not driven by thin data.
- Tax Rate has the smallest coverage, but even there the sample is still large enough to make precise cross-sectional estimation feasible. That makes the over-rejection result more striking, since it survives in large samples.

5.3 Figure 2: Significance rates based on cross-sectional standard errors

Read:

- At the 1% level with default OLS standard errors, significance rates are huge: 49.60% for Size, 43.68% for M/B , 47.82% for Cash/AT, 31.52% for Tax Rate, and even 24.67% for Debt/AT. NY HQ is the least problematic but still significant on 6.86% of days.
- At the 5% level with default OLS standard errors, the numbers are even larger: 61.00% for Size, 57.69% for Cash/AT, 55.51% for M/B , and 42.71% for Tax Rate.
- Robust and clustered standard errors reduce rejection rates somewhat, but not enough. At the 1% level with FF49 clustering, Size still rejects on 33.70% of days, Cash/AT on 28.36%, and Tax Rate on 15.34%.
- The variables with the largest false-positive problem are precisely those that capture broad economic exposure – Size, M/B , and Cash/AT. NY HQ is less exposed to routine news, so it generates lower ordinary-day slopes.
- This figure establishes the practical magnitude of the problem. Cross-sectional standard errors are not just conceptually wrong; they are empirically disastrous.

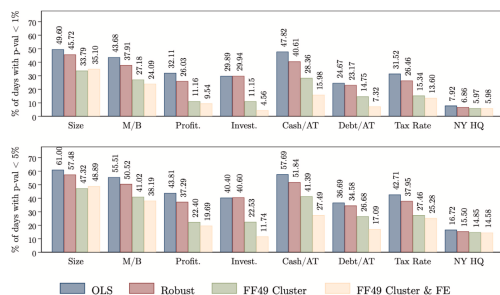


Fig. 2. Significance rates based on cross-sectional standard errors. This figure depicts statistical significance rates from univariate cross-sectional OLS regressions of 1-day returns on each of eight characteristics. The bars show the percentage of days for which the coefficient from the regression is statistically significant based on default standard errors, White (1980) robust standard errors, Fama-French 49-category industry clustered standard errors, and industry clustered standard errors where we also include industry fixed effects. The top panel shows rates of significance at the 1% significance level, while the bottom panel shows rates of significance at the 5% significance level. Our sample is the 7,811 trading days in 1991 through 2021.

Table 2
Significance rates based on cross-sectional standard errors.

Panel A: Different Standard Errors									
Return		1% Significance Level				5% Significance Level			
		Univariate		Multivariate		Univariate		Multivariate	
		1d	5d	1d	5d	1d	5d	1d	5d
Unadj.	Default SE	33.4%	40.8%	26.1%	33.5%	44.3%	51.3%	37.4%	44.4%
Unadj.	Robust SE	29.6%	37.1%	22.5%	29.7%	40.7%	47.9%	33.8%	41.0%
Unadj.	FF49 Cluster	18.5%	22.9%	16.3%	21.6%	30.5%	35.2%	28.0%	33.7%
Unadj.	FF49 Cl & FE	14.5%	19.6%	13.2%	20.0%	25.4%	30.9%	23.7%	31.6%

Panel B: Alternative Returns									
Return		1% Significance Level				5% Significance Level			
		Univariate		Multivariate		Univariate		Multivariate	
		1d	5d	1d	5d	1d	5d	1d	5d
CAPM	FF49 Cl & FE	17.7%	21.4%	12.2%	17.1%	29.8%	33.7%	22.4%	28.1%
FF3	FF49 Cl & FE	14.0%	16.6%	10.1%	14.7%	25.2%	28.3%	20.0%	25.4%
FF4	FF49 Cl & FE	13.2%	16.0%	9.5%	13.9%	24.2%	27.5%	19.2%	24.5%
LogRet	FF49 Cl & FE	18.5%	22.9%	13.2%	18.1%	30.5%	35.0%	23.7%	29.5%

This table presents statistical significance rates for coefficients from cross-sectional OLS regressions of short-term stock returns on each of eight firm characteristics based on cross-sectional standard errors. Each cell reports the percentage of all one- or five-day periods in which the regression coefficient is statistically significant, averaged across the eight characteristics. Each row reports the percentages for different combinations of 1% and 5% significance, univariate and multivariate regressions (where all eight characteristics are included as explanatory variables), and one-day and five-day return periods. Panel A presents rates of significance based on default standard errors, White (1980) robust standard errors, and standard errors clustered by Fama-French 49-category industry, with the fourth row based on regressions that also include Fama-French 49-category industry fixed effects. Panel B presents rates of significance for alternative return measures, including CAPM, Fama-French 3-factor, and Fama-French 4-factor adjusted returns and logged returns ($\log(1 + \text{return})$). Our sample is the 7,811 trading days in 1991 through 2021.

5.4 Table 2: Alternative specifications for cross-sectional tests

Read:

- Five-day windows make the problem worse. Under default standard errors, average one-day univariate rejection at the 1% level is 33.4%, while the five-day number is 40.8%. At the 5% level the corresponding averages are 44.3% and 51.3%.
- Multivariate regressions help because control variables absorb overlapping exposure to routine news. For example, with default standard errors at the 1% level, the average one-day rejection rate falls from 33.4% in univariate regressions to 26.1% in multivariate regressions.
- Industry clustering plus industry fixed effects helps more, but still leaves very large over-rejection: for one-day univariate regressions the average 1% rejection rate is 14.5%, and the average 5% rejection rate is 25.4%.
- Factor-adjusted returns are not enough either. With FF4-adjusted returns plus FF49 clustering and industry fixed effects, the one-day univariate 1% rejection rate is still 13.2%, and the 5% rate is 24.2%.
- Economic lesson: better controls and better return definitions reduce the problem but do not solve the null-hypothesis mismatch.

5.5 Table 3: Alternative clustering choices

Read:

- The problem is robust to many clustering dimensions. With one-day univariate regressions at the 1% level, rejection rates are 22.2% with SIC4 clustering, 21.3% with FF30 clustering, 30.2% with state clustering, and 20.6% with size-by- M/B bin clustering.
- State clustering performs especially poorly, with a 42.3% average rejection rate at the 5% level for one-day univariate regressions.
- Two-way clustering does not rescue inference either: FF49 plus state clustering still gives a 20.7% average rejection rate at the 1% level and 32.2% at the 5% level for one-day univariate regressions.
- This table drives home that the issue is not “choosing the right cluster.” The missing uncertainty is not a conventional residual-correlation problem.

5.6 Figure 3: Significance rates based on TS-OLS and TS-GLS

Read:

- The CDF-based versions of both TS-OLS and TS-GLS bring rejection rates close to nominal size. At the 1% level, the CDF rates are roughly around 1%–1.6% depending on the characteristic; at the 5% level they sit around 5.3%–5.8%.
- The t -test versions still over-reject slightly. At the 1% level, t -based rates are around 1.8%–2.3%; at the 5% level they are around 5.5%–6.1%.

Table 3
Significance rates for alternative cross-sectional standard errors.
Panel A: Alternative Clustering

Return	Specification	1% Significance Level				5% Significance Level			
		Univariate 1d	Univariate 5d	Multivariate 1d	Multivariate 5d	Univariate 1d	Univariate 5d	Multivariate 1d	Multivariate 5d
Unadj.	SIC4 Cluster	22.2%	27.8%	18.7%	24.6%	33.9%	39.6%	30.0%	36.4%
Unadj.	FF49 Cluster	18.5%	22.9%	16.3%	21.6%	30.5%	35.2%	28.0%	33.7%
Unadj.	FF30 Cluster	21.3%	25.7%	18.5%	23.7%	30.7%	34.9%	28.3%	33.6%
Unadj.	State Cluster	30.2%	36.2%	25.7%	31.9%	42.3%	48.5%	37.6%	44.4%
Unadj.	5 Size x 5 MB	20.6%	25.7%	17.0%	23.0%	32.7%	38.6%	29.0%	35.5%
Unadj.	... x SGP	24.2%	30.4%	19.7%	26.4%	35.9%	42.1%	31.1%	38.0%

Return	Specification	1% Significance Level				5% Significance Level			
		Univariate 1d	Univariate 5d	Multivariate 1d	Multivariate 5d	Univariate 1d	Univariate 5d	Multivariate 1d	Multivariate 5d
Unadj.	FF49 & State	20.7%	24.6%	18.8%	23.5%	32.2%	36.2%	30.0%	35.4%
Unadj.	FF30 & State	20.1%	23.7%	16.7%	21.0%	31.7%	35.5%	27.6%	32.6%

This table presents mean statistical significance rates for coefficients from cross-sectional OLS regressions of short-term stock returns on each of eight firm characteristics based on alternative cross-sectional standard errors. Each cell reports the percentage of all 1- or 5-day periods from 1991 through 2021 in which the regression coefficient is statistically significant. Each row reports the percentages for different combinations of 1% and 5% significance, univariate and multivariate regressions (where all eight characteristics are included as explanatory variables), and one-day and five-day return periods. Panel A presents rates of significance based on standard errors clustered at different industry levels, at the state level, at the level of size quintile by book-to-market quintile bin, and at the level of size quintile by book-to-market quintile by gross profitability quintile bin. Panel B presents rates of significance based on standard errors clustered by both industry and size.

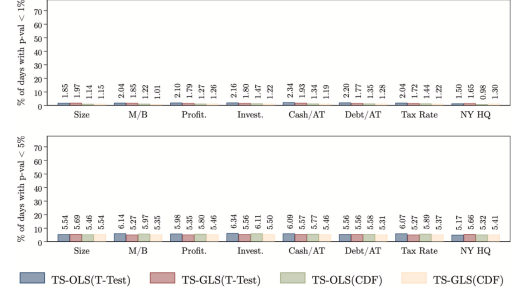


Fig. 3. Significance rates based on TS-OLS and TS-GLS.
This figure depicts statistical significance rates for the relationship between one-day returns and each of eight characteristics using the time-series OLS (TS-OLS) and time-series GLS (TS-GLS) approaches described in Section 2.4. We use 199 pre-event periods and, for each covariance matrix used in TS-GLS, 100 principal components extracted from the 200 preceding trading days. The bars show the percentage of days for which the coefficient from the regression is statistically significant based on t -statistics using the time-series of coefficients as standard errors (T-Test) or based on a p value from comparison to the empirical time-series cumulative distribution function of coefficients (CDF). The top panel shows rates of significance at the 1% significance level, while the bottom panel shows rates of significance at the 5% significance level. Our sample is the 7,811 trading days in 1991 through 2021.

- The gap between the t -test bars and the CDF bars is direct evidence of fat tails in the empirical slope distribution.
- Size control is therefore achieved not merely by going from cross-sectional to time-series inference, but by using the empirical distribution rather than imposing a thin-tailed parametric approximation.

5.7 Figure 4 and Table 4: Detection rates in artificial event periods

Read:

- The paper adds artificial event effects of 25 or 50 basis points per one-standard-deviation increase in each characteristic and asks how often the methods detect them.
- For one-day 25 bp effects at the 5% level, TS-GLS strongly dominates TS-OLS: Size rises from 18.68% to 33.54%, M/B from 29.02% to 65.42%, Profit from 29.05% to 63.76%, Invest from 39.15% to 77.26%, Debt/AT from 36.53% to 81.30%, and Tax Rate from 69.93% to 98.78%.
- For one-day 50 bp effects, TS-GLS nearly always dominates by economically large margins: the average detection rate rises from 72.90% to 95.46% at the 5% level.
- Table 4 summarizes the average gains. At the 1% level, a one-day 25 bp effect is detected 20.76% of the time under TS-OLS and 49.05% under TS-GLS (ratio 2.36). A five-day 25 bp effect is detected only 5.20% of the time under TS-OLS but 14.89% under TS-GLS (ratio 2.86).
- At the 5% level, the corresponding averages are 40.93% versus 71.82% for one-day 25 bp events, and 35.19% versus 61.24% for five-day 50 bp events.
- Main message: correcting size is not enough; power matters. TS-GLS materially improves the odds of detecting realistic event effects.

5.8 Table 5: Robustness of TS-GLS to L and K

Read:

- The power gains are robust to the number of pre-event days and principal components.

Table 4

Detection rates in artificial event periods: TS-OLS vs TS-GLS.

Panel A: 1% Significance Level					
Period	Effect (bp)	TS-OLS	TS-GLS	Difference	Ratio
1 day	25	20.76%	49.05%	28.29%	2.36
1 day	50	53.20%	85.94%	32.74%	1.62
5 day	25	5.20%	14.89%	9.69%	2.86
5 day	50	19.03%	42.34%	23.31%	2.22

Panel B: 5% Significance Level					
Period	Effect (bp)	TS-OLS	TS-GLS	Difference	Ratio
1 day	25	40.93%	71.82%	30.89%	1.75
1 day	50	72.90%	95.46%	22.56%	1.31
5 day	25	15.29%	30.08%	14.79%	1.97
5 day	50	35.19%	61.24%	26.05%	1.74

This table presents statistical significance rates for the relationship between short-term (one- or five-day) returns and each of eight characteristics using the time-series OLS (TS-OLS) and time-series GLS (TS-GLS) approaches described in Section 2.4. For each one- or five-day period, we create an artificial event by adding either 25 or 50 bp to a firm's return for each one standard deviation increase in the characteristic in question, spreading the effect equally across days in the case of five-day periods. We use 199 pre-event periods (with no added relationship) and, for each covariance matrix used in TS-GLS, 100 principal components extracted from the 200 preceding trading days. Each row reports the percentage of all artificial event periods in which the estimated event-period relationship is statistically significant using TS-OLS and TS-GLS based on a p value from comparison to the empirical time-series cumulative distribution function of coefficients as well as the difference and ratio of these percentages. Different rows report these percentages for different combinations of event-period length (one- or five-day) and size of added return per one standard deviation increase in a characteristic (25 or 50 bp). The top panel shows significance rates at the 1% level, while the bottom panel shows significance rates at the 5% level. Our sample is the 7,811 trading days in 1991 through 2021.

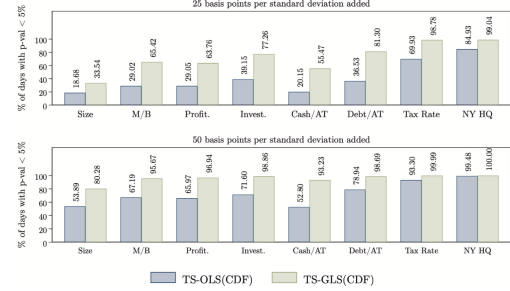


Fig. 4. Detection rates in artificial event periods: TS-OLS vs TS-GLS. This figure depicts statistical significance rates for the relationship between one-day returns and each of eight characteristics using the time-series OLS (TS-OLS) and time-series GLS (TS-GLS) approaches described in Section 2.4. For each day, we create an artificial event by adding either 25 or 50 bp to a firm's return for each one standard deviation increase in the corresponding characteristic on the x axis. We use 199 pre-event days (with no added relationship) and, for each covariance matrix used in TS-GLS, 100 principal components extracted from the 200 preceding trading days. The bars show the percentage of days for which the coefficient from the regression is statistically significant at the 5% level based on a comparison to the empirical time-series cumulative distribution function of coefficients (CDF). The top panel shows results for a 25 bp added relationship, while the bottom panel shows results for a 50 bp added relationship. Our sample is the 7,811 trading days in 1991 through 2021.

- At the 5% level with a one-day 25 bp artificial effect, detection ranges only from 70.60% to 72.49% across $L \in \{99, 199, 299\}$ and $K \in \{50, 100, 150\}$.
- For a one-day 50 bp effect at the 5% level, detection stays between 95.30% and 95.57% across the same grid.
- For five-day 50 bp effects at the 5% level, detection ranges from 57.34% to 64.34%.
- This is reassuring because it means the paper's substantive recommendation does not depend on a knife-edge covariance-specification choice.

5.9 Figure 5: Explanatory power of principal components

Read:

- The figure plots how much of the daily cross-sectional variance of returns is explained by the first 1, 5, 25, and 100 PCs.
- The first 5 PCs explain less than 30% of return variation in most years. Even the first 25 PCs explain less than 50% in most years.
- Moving from 25 to 100 PCs nearly doubles explained variation. The 100-PC specification explains roughly 60% to 80% depending on the year.
- Fama-French 49 industry portfolios explain only about as much as the first PC. That is why industry clustering is too crude for the problem at hand.
- Interpretation: return comovements are genuinely high-dimensional, so a rich PCA-based covariance estimate is not overkill; it is necessary.

Table 5
TS-GLS detection rates in artificial event periods: Robustness.

Panel A: 1% Significance Level							
Effect (bp)	L	1 day			5 day		
		$K = 50$	$K = 100$	$K = 150$	$K = 50$	$K = 100$	$K = 150$
25	99	47.87%	48.91%	48.02%	16.12%	17.73%	17.62%
25	199	48.06%	49.05%	48.58%	13.46%	14.89%	14.71%
25	299	48.24%	49.40%	48.47%	12.54%	13.75%	13.69%
50	99	83.87%	84.74%	84.24%	43.55%	47.25%	47.77%
50	199	85.30%	85.94%	85.40%	39.04%	42.34%	42.84%
50	299	85.06%	85.64%	85.43%	37.83%	40.55%	41.12%

Panel B: 5% Significance Level							
Effect (bp)	L	1 day			5 day		
		$K = 50$	$K = 100$	$K = 150$	$K = 50$	$K = 100$	$K = 150$
25	99	71.68%	72.49%	71.75%	30.01%	31.94%	32.37%
25	199	71.01%	71.82%	71.13%	28.31%	30.08%	30.48%
25	299	70.60%	71.34%	70.85%	27.74%	29.31%	29.54%
50	99	95.32%	95.51%	95.43%	60.05%	63.49%	64.34%
50	199	95.30%	95.46%	95.35%	57.94%	61.24%	62.23%
50	299	95.38%	95.57%	95.41%	57.34%	60.41%	61.38%

This table presents mean rates with which the time-series GLS (TS-GLS) approach to significance testing described in Section 2.4 successfully detects relationships between one-day returns and each of eight characteristics added to create artificial event days for different numbers of pre-event days (L) and principal components (K) used to construct covariance matrices. We use the preceding 200 trading days to compute principal components. Each row reports the percentage of all artificial event periods from 1991 to 2021 in which the estimated event-period relationship is statistically significant for different numbers of principal components. Different rows report these percentages for different combinations of number of pre-event days and size of added return per one standard deviation increase in a characteristic (25 or 50 bp). The top panel shows significance rates at the 1% level, while the bottom panel shows significance rates at the 5% level.

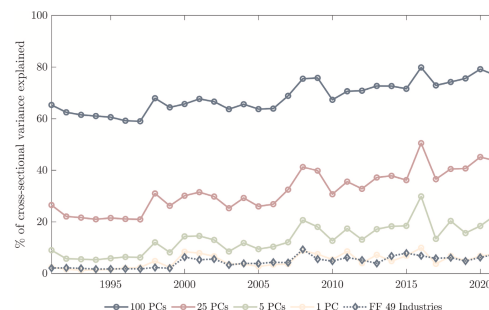


Fig. 5. Explanatory power of principal components of daily returns.
This figure plots the average percent of the daily cross-sectional variance of returns that the first K principal components (PCs) combined explain by calendar year, for $K = 1, 5, 25$, and 100 . It also shows the average percent explained by Fama-French 49-category industry portfolios combined for comparison. Our sample is the 7,811 trading days in 1991 through 2021.

5.10 Table 6 and Figure 6: What the principal components are picking up

Read:

- Table 6 maps the first five PCs in selected years into interpretable factor portfolios.
- In 2008, PC1 looks like the market (88% correlation with $MktRf$), while PC3 looks like a crisis-finance factor, with 60% correlation with Finance, 46% with HML , and 45% with UMD .
- In 2010, only PC2 strongly resembles the market (95% R^2 with the factor menu), suggesting a relatively quiescent environment.
- In 2020, PC4 looks like a COVID factor (52% R^2 and 39% correlation with the COVID portfolio), while PC2 lines up with Finance.
- In 2021, PC2 is overwhelmingly a meme-stock factor, with 93% correlation with the Memes portfolio. PC4 also correlates with Tech and Finance.
- Figure 6 broadens the picture over time. Finance spikes in importance in 2008, Memes in 2021, and even standard factors such as SMB vary a lot in their explanatory power over time.
- Overall lesson: the relevant covariance structure is episode-specific. That is exactly why pre-specified low-dimensional factor models do not solve the inference problem well.

5.11 Appendix Table A.1: Survey of recent event-study practice

Read:

- The appendix surveys 81 papers with a combined 16,070 Google Scholar citations.
- 49% use CAPM-adjusted returns, 25% simple returns, 20% Fama-French adjusted returns, 5% industry-adjusted returns, and 1% log returns.

Table 6

Evolving correlations with principal components.

Panel A: 2008					
PC	'Mkt'	?	'Crisis'	?	?
1	2	3	4	5	
% x-sectional var. explained	5.9%	3.6%	3.0%	2.7%	2.1%
$ \rho(PC_1, MktRf) $	88%	4%	13%	2%	3%
$ \rho(PC_1, SMB) $	28%	12%	41%	14%	20%
$ \rho(PC_1, HML) $	4%	4%	46%	0%	30%
$ \rho(PC_1, UMD) $	3%	8%	45%	2%	36%
$ \rho(PC_1, Tech) $	7%	9%	5%	3%	6%
$ \rho(PC_1, Finance) $	6%	4%	60%	4%	35%
$ \rho(PC_1, Covid) $	5%	9%	27%	1%	2%
$ \rho(PC_1, Memes) $	2%	4%	8%	6%	2%
R^2	88%	5%	55%	3%	19%

Panel B: 2010					
PC	?	'Mkt'	?	?	?
1	2	3	4	5	
% x-sectional var. explained	5.2%	4.0%	1.6%	1.4%	1.2%
$ \rho(PC_1, MktRf) $	1%	92%	10%	8%	5%
$ \rho(PC_1, SMB) $	1%	30%	12%	23%	31%
$ \rho(PC_1, HML) $	3%	3%	1%	26%	13%
$ \rho(PC_1, UMD) $	3%	10%	15%	25%	22%
$ \rho(PC_1, Tech) $	2%	0%	7%	6%	2%
$ \rho(PC_1, Finance) $	9%	5%	7%	2%	6%
$ \rho(PC_1, Covid) $	1%	0%	7%	22%	25%
$ \rho(PC_1, Memes) $	1%	1%	14%	3%	1%
R^2	1%	95%	10%	21%	22%

Panel C: 2020					
PC	'Mkt'	'Fin'	?	'Covid'	?
1	2	3	4	5	
% x-sectional var. explained	6.2%	4.7%	2.8%	2.5%	2.2%
$ \rho(PC_1, MktRf) $	54%	24%	4%	7%	7%
$ \rho(PC_1, SMB) $	17%	43%	1%	32%	13%
$ \rho(PC_1, HML) $	36%	53%	9%	39%	0%
$ \rho(PC_1, UMD) $	29%	55%	1%	45%	8%
$ \rho(PC_1, Tech) $	21%	34%	9%	19%	15%
$ \rho(PC_1, Finance) $	39%	49%	8%	19%	10%
$ \rho(PC_1, Covid) $	24%	16%	11%	39%	19%
$ \rho(PC_1, Memes) $	25%	32%	8%	9%	26%
R^2	49%	45%	6%	52%	16%

Panel D: 2021					
PC	'Quant'	'Memes'	?	'Tech'	?
1	2	3	4	5	
% x-sectional var. explained	5.4%	5.1%	3.1%	2.3%	2.2%
$ \rho(PC_1, MktRf) $	36%	11%	0%	17%	27%
$ \rho(PC_1, SMB) $	36%	44%	2%	27%	29%
$ \rho(PC_1, HML) $	45%	2%	37%	45%	48%
$ \rho(PC_1, UMD) $	35%	21%	7%	8%	2%
$ \rho(PC_1, Tech) $	9%	21%	17%	36%	30%
$ \rho(PC_1, Finance) $	14%	5%	27%	46%	34%
$ \rho(PC_1, Covid) $	14%	9%	14%	2%	4%
$ \rho(PC_1, Memes) $	8%	93%	1%	4%	6%
R^2	59%	89%	15%	33%	40%

This table presents the correlations between portfolios representing the first five principal components of individual stock returns in a calendar year and a variety of other factors constructed from stock returns. The first four are the market ($MktRf$), size (SMB), value (HML) and momentum (UMD) factors. The second four are period-specific factors: $Tech$, the equal-weighted average return of Software, Hardware, and Chips Fama-French 49 industries; $Finance$, the average return of Banks, Real Estate, and Finance industries; $Covid$, the average return of Meals, Healthcare, and Drugs industries; and $Memes$, the average return of whichever subset of GameStop (GME), AMC (AMC), Bed Bath and Beyond (BBBY), and Blackberry (BB) stocks is available to trade on each day. We orthogonalize the non- $MktRf$ factors with respect to $MktRf$ in each calendar year. Panel A presents results for 2008, Panel B for 2010, Panel C for 2020, and Panel D for 2021.

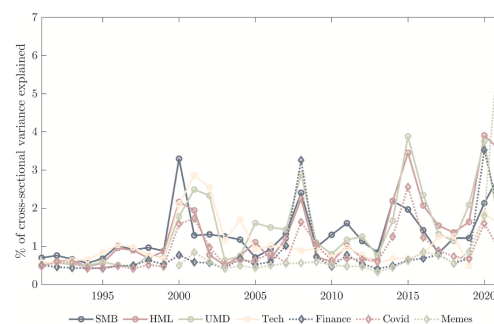


Fig. 6. Explanatory power of example factors for daily returns.

This figure plots the average percent of the daily cross-sectional variance of returns that seven example factors explain by calendar year. The factors are size (SMB), value (HML) and momentum (UMD) factors, as collected from Ken French's data library; $Tech$, the equal-weighted average return of Software, Hardware, and Chips Fama-French 49 industries; $Finance$, the average return of Banks, Real Estate, and Finance industries; $Covid$, the average return of Meals, Healthcare, and Drugs industries; and $Memes$, the average return of whichever subset of GameStop (GME), AMC (AMC), Bed Bath and Beyond (BBBY), and Blackberry (BB) stocks is available to trade on each day. All factors are orthogonalized with respect to the excess market return ($MktRf$) in each calendar year. Our sample is the 7,811 trading days in 1991 through 2021.

- 64% use simple or robust standard errors, 20% use industry clustering, 5% geography clustering, 6% time-series methods, and 5% other approaches.
- The survey is important because it shows that the paper is not fixing a marginal technicality. It is correcting a common convention in the literature.

Table A.1
Survey of cross-sectional event studies involving a quasi-experimental event.

Panel A: Summary Statistics					
Total Papers	81				
Total Google Cites	16,079				
Return Types	Standard Errors				
Simple	20	25%	Simple	52	64%
CAPM Adjusted	40	49%	Industry Cluster	16	20%
FF Adjusted	16	20%	Geography Cluster	4	5%
Industry Adjusted	4	5%	Time-Series	5	6%
Log	1	1%	Other	4	5%
Fixed Effects					
None	45	56%			
Industry	30	37%			
Geography	1	1%			
Industry & Geography	5	6%			

Panel B: Survey of papers					
Paper	Journal	Google Cites	Return Type	Fixed Effects	Standard Errors
Abadie et al. (2003)	JBF	36	Simple	None	Simple/Robust
Aceronigle et al. (2016)	JFE	844	CAPM Adj.	None	Simple/Robust
Aceronigle et al. (2016)	RFS	540	Simple	Industry	Simple/Robust
Acker et al. (2018)	JBF	11	Simple	Industry	Simple/Robust
Alex and Apal (2011)	JF	228	CAPM Adj.	None	Simple/Robust
Akylidis (2019)	JFE	19	CAPM Adj.	Industry	Time-Series
Albuquerque et al. (2020)	JCF	14	CAPM Adj.	Industry	Ind. Cluster
Albuquerque et al. (2020)	RCFS	1306	CAPM Adj.	Industry	Simple/Robust
Amihud and Shyamov (2017)	JFE	76	FF Adj.	Industry	Simple/Robust
Anow et al. (2022)	RFS	151	CAPM Adj.	Ind. & Geog.	Simple/Robust
Andrade and Chiachabata (2018)	RFS	36	FF Adj.	Industry	Ind. Cluster
Audrey et al. (2020)	JBF	20	CAPM Adj.	None	Simple/Robust
Angen et Li (2019)	JCF	39	Simple	None	Simple/Robust
Bae et al. (2021)	JCF	641	Simple	Industry	Simple/Robust
Berkowitz et al. (2013)	JFE	213	CAPM Adj.	Industry	Simple/Robust
Beaudouin et al. (2017)	JCF	174	Simple	Ind. & Geog.	Geo. Cluster
Bijay et al. (2017)	JAE	73	FF Adj.	Industry	Ind. Cluster
Bisak et al. (2022)	JF	38	CAPM Adj.	Industry	Simple/Robust
Black et al. (2019)	JBF	148	Ind. Cluster	None	Simple/Robust
Bongali et al. (2013)	JBF	140	CAPM Adj.	None	Simple/Robust
Bortner et al. (2018)	RFS	272	FF Adj.	Industry	Time-Series
Brown and Huang (2020)	JFE	198	CAPM Adj.	Industry	Ind. Cluster

Carboel et al. (2017)	JBF	71	CAPM Adj.	None	Simple/Robust
Chen and Kwik (2017)	JBF	74	Simple	Industry	Simple/Robust
Chang et al. (2019)	JBF	348	CAPM Adj.	None	Simple/Robust
Chen et al. (2018)	JAE	1384	FF Adj.	Industry	Simple/Robust
Chen et al. (2019)	JCF	63	FF Adj.	None	Simple/Robust
Child et al. (2011)	JFE	67	CAPM Adj.	Industry	Ind. Cluster
Choy et al. (2017)	JCF	50	CAPM Adj.	None	Other
Chung et al. (2020)	JCF	41	CAPM Adj.	None	Simple/Robust
Cohn et al. (2016)	JF	107	Simple	None	Time-Series
Coutins et al. (2020)	MS	91	CAPM Adj.	None	Ind. Cluster
Dodgale and Dyck (2013)	JF	179	Simple	None	Ind. Cluster
Fahlenbrach et al. (2021)	RFS	608	Simple	Industry	Simple/Robust
Feng et al. (2022)	JBF	59	FF Adj.	Industry	Simple/Robust
Flannan et al. (2014)	RFS	188	FF Adj.	Industry	Simple/Robust
Foley-Fisher et al. (2014)	JFE	179	CAPM Adj.	None	Ind. Cluster
Fontana (2021)	JBF	50	CAPM Adj.	None	Simple/Robust
Go et al. (2018)	JBF	71	Industry Adj.	None	Simple/Robust
Gent and Pridemore (2021)	JCF	280	Simple	Industry	Simple/Robust
Gilje and Talland (2017)	RFS	133	Simple	None	Simple/Robust
Gómez-Grau and Gómez (2023)	RFS	16	Simple	Industry	Simple/Robust
Groene et al. (2020)	JCF	228	CAPM Adj.	Industry	Ind. Cluster
Groval et al. (2019)	MS	566	Simple	None	Geo. Cluster
Hackbush et al. (2013)	RFS	95	Simple	None	Simple/Robust
Hill et al. (2019)	JBF	110	Simple	Industry	Simple/Robust
Huang et al. (2020)	JCF	19	CAPM Adj.	Geography	Simple/Robust
Igan et al. (2022)	JBF	55	Log	None	Geo. Cluster
Jain et al. (2019)	JBF	11	Simple	None	Simple/Robust
Kalchauer et al. (2020)	JBF	37	FF Adj.	Industry	Other
Kim and Kim (2022)	JCF	2	CAPM Adj.	None	Time-Series
Kolstad and Yang (2022)	MS	0	FF Adj.	None	None
Rundlo (2022)	JFQA	1	FF Adj.	Ind. & Geog.	Ind. Cluster

(continued on next page)

Table A.1 (continued).

Lewis and Veredemans (2014)	JCF	19	CAPM Adj.	Industry	Simple/Robust
Licht et al. (2018)	JFE	41	CAPM Adj.	None	Other CLS
Liu et al. (2019)	JAE	42	Industry Adj.	None	Simple/Robust
Liu et al. (2017)	JF	3837	Simple	Industry	Simple/Robust
Liu et al. (2017)	JFE	305	CAPM Adj.	None	Ind. Cluster
Liu et al. (2021)	JBF	43	CAPM Adj.	None	Ind. Cluster
Liu et al. (2022)	JBF	19	CAPM Adj.	None	Simple/Robust
Lippenberger (2018)	JBF	67	CAPM Adj.	None	Other CLS
March (2022)	JBF	5	FF Adj.	None	Simple/Robust
May (2014)	JCF	18	FF Adj.	None	Simple/Robust
Moon and Schooner (2022)	JFE	15	Simple	None	Simple/Robust
Nichols et al. (2022)	JAE	18	CAPM Adj.	None	Simple/Robust
Neukirchen et al. (2020)	JBF	20	CAPM Adj.	Ind. & Geog.	Geo. Cluster
Neyens and Pilon (2020)	JCF	273	CAPM Adj.	None	Simple/Robust
O'Donovan et al. (2019)	RFS	239	CAPM Adj.	None	Time-Series
Pagan et al. (2022)	JFE	265	CAPM Adj.	None	Ind. Cluster
Ranelli et al. (2021)	RCFS	131	CAPM Adj.	Industry	Simple/Robust
Ranelli et al. (2021)	JCF	91	CAPM Adj.	Ind. & Geog.	Simple/Robust
Rayner and Siro-Roman (2020)	JCF	9	CAPM Adj.	None	Simple/Robust
Schooner (2019)	JF	325	Simple	Industry	Simple/Robust
Shao and Tang (2022)	RuF	139	FF Adj.	Industry	Ind. Cluster
Stall (2022)	RuF	2	Industry Adj.	None	Simple/Robust
van Bellen (2014)	JBF	7	Simple	Industry	Simple/Robust
Wagner et al. (2018)	JFE	337	Simple	Industry	Simple/Robust
Wang and Chen (2018)	JBF	120	FF Adj.	None	Simple/Robust
Wilson (2022)	JCF	5	FF Adj.	None	Simple/Robust
Zouane (2017)	RFS	217	CAPM Adj.	Industry	Ind. Cluster
Zhang (2012)	JBF	18	CAPM Adj.	None	Simple/Robust

This table lists and summarizes 81 papers published in top finance and accounting journals (JAE, JAR, JBF, JCF, JF, JFE, JFE, JFQA, MS, RCFS, RFS, RuF) from 2014-2023 using the cross-sectional event studies methodology around a quasi-experimental event. Panel A shows the total Google Scholar citation count across all papers, and provides paper counts and fractions of the total paper count for each category of return type, fixed effects, and standard errors. The five categories of returns used are simple or raw returns ("Simple"), CAPM or market adjusted ("CAPM Adj."), Fama French or factor adjusted ("FF Adj."), industry adjusted ("Industry Adj."), Log ("Log"). The four categories of fixed effects used are None, Industry, Geography, and Industry & Geography. The five categories of standard error calculation are standard OLS or within (1980) robust ("Simple/Robust"), clustered by industry ("Ind. Cluster"), clustered by Geography ("Geo. Cluster"), time-series or portfolio-based ("Time-Series"), and bootstrapped or other cross-sectional approach ("Other Cluster"). The number of papers and the number of published papers are listed in the number of Google Scholar citations as of January 2025, the return type, which fixed effects are used, and how standard errors are calculated.

5.12 Appendix Figures C.1 and C.2: Choosing the number of PCs

Read:

- Figure C.1 shows that the volatility of an out-of-sample minimum-variance portfolio falls sharply when the first several PCs are added, then becomes nearly flat over a large interior region before rising near 200 PCs. This suggests the useful covariance information is front-loaded but not exhausted after only a handful of components.
- Figure C.2 shows that the ratio of GLS to OLS coefficient volatility falls sharply with the first 20 or so PCs, then is nearly flat from about 40 to 190 PCs for all eight characteristics.
- Practical interpretation: $K = 100$ is a safe middle ground. It is rich enough to capture most covariance benefits without getting close to the overfitting region.

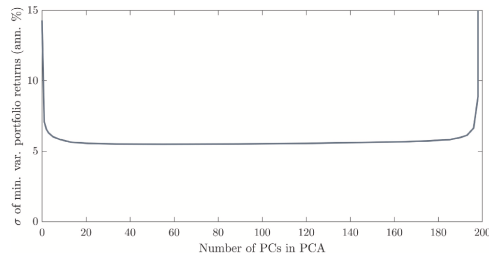


Fig. C.1. Minimum variance portfolio and # of principal components.

This figure depicts the average annualized standard deviation of daily returns for a minimum-variance portfolio of US equities as a function of the number of principal components (PCs), K , used to form an out-of-sample forecast for the covariance matrix of returns. Our covariance matrix forecasts are constructed using the first K PCs from implementing principal component analysis on a balanced panel of daily returns over the prior 200 trading days, assuming that, other than via these PCs, each stock's return is uncorrelated. See Section 2.4 for more details. Our sample is the 7,811 trading days in 1991 through 2021.

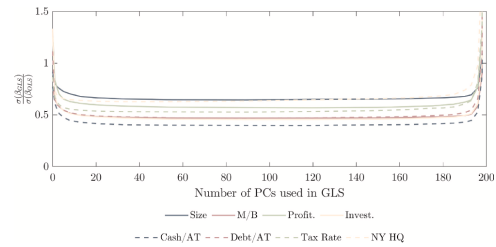


Fig. C.2. GLS/OLS coefficient volatility and # of principal components.

This figure plots the ratio of the time-series standard deviations of cross-sectional GLS and OLS coefficients from cross-sectional regressions of one-day returns on each of eight characteristics against the number of PCs we use to estimate the covariance matrix for the GLS regressions. See Section 2.4 for more details. Our sample is the 7,811 trading days in 1991 through 2021.

6 Short summary

- **Conclusion.** Standard cross-sectional event-day standard errors dramatically overstate the evidence for differential event effects because they ignore the time-series volatility of ordinary-

day return-characteristic slopes.

- **Intuition.** The focal event arrives on a day that also contains routine news. The right comparison is therefore not “event-day slope versus zero” but “event-day slope versus the distribution of ordinary-day slopes.”
- **Empirical lesson.** Use a pre-event time-series benchmark, prefer empirical-CDF inference to a parametric t -test, and use the paper’s PCA-based TS-GLS method when power matters.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- **Main finding.** Conventional event-day cross-sectional regressions produce far too many false positives for reliable causal inference in return-characteristic event studies.
- **Quantitative magnitude.** At the 1% significance level, default cross-sectional tests reject on 33.4% of ordinary days on average; even FF49 industry clustering leaves an 18.5% rejection rate.
- **Correct inferential logic.** The relevant null is whether the event-day slope is unusual relative to pre-event slopes, not whether the event-day slope is zero in a single cross section.
- **Best-performing procedure.** The empirical-CDF version of the time-series approach fixes size, and the PCA-based TS-GLS implementation adds major power gains over TS-OLS.
- **Broader econometric implication.** Clustering, factor adjustment, and alternative window definitions are not enough when the missing uncertainty comes from slope-level variation rather than residual correlation.
- **Economic implication.** Return comovements are complex and unstable across time. Inference procedures that hard-code a small set of factors or a simple clustering structure will systematically miss the relevant covariance patterns.

7.2 Economic intuition

- On any given day, many news items arrive. Firms that share a characteristic such as size, investment, or cash holdings often react similarly to that news. So the event-day cross-sectional slope is noisy in a structural sense, not just a sampling sense.
- Clustering standard errors is aimed at correlated residuals, but the confounding-events problem is about the event-day coefficient itself being shifted by background news.
- The PCA-based covariance matrix improves power because it learns the actual dimensions along which firms co-move in the recent past, and those dimensions are often richer than industry or standard factor partitions.

7.3 Takeaway

This paper’s main contribution is to replace a misleading event-study benchmark with the right one. A researcher should only treat a return-characteristic relationship around an event as evidence

of a differential treatment effect if that relationship stands out relative to the same relationship on ordinary nearby days. Once the question is posed that way, the empirical CDF delivers valid size, and the paper's TS-GLS procedure delivers much better power. For anyone doing cross-sectional return event studies, this is a direct change in applied practice rather than a minor technical refinement.